century, ecologists devoted more attention to the interrelatedness of all organisms within an ecosystem, exemplified by Russian scientist Vladimir Vernadsky’s concept of the biosphere.

While studying the vegetation growing on sand dunes along the shore of Lake Michigan in the 1890s, American botanist Henry Chandler Cowles realized that there was a succession of plant species, with “pioneer” plants being replaced by others, which were in turn themselves supplanted. Fellow American Frederic Clements used the term “climax community” to describe the endpoint of this succession. In 1916, he proposed that global vegetation patterns could be thought of as “formations,” or large communities of plants —and the organisms that depended on them—which reflected the regional climate. In relatively wet, temperate regions, for example, deciduous forest may dominate, but grassland tends to dominate in drier, more temperate areas. Clements argued that these climax communities were bound together and could be thought of as single, complex organisms.

Clements was soon challenged by American botanist Henry Gleason, who agreed that plant communities could be mapped, but argued that since individual plant species have no common purpose, the idea of integrated communities was invalid. His view found support in the 1950s, in the field studies of Robert Whittaker and the numerical research of John Curtis.

In 1967, American ecologist Richard Root proposed the idea of the “guild,” a group of organisms—closely related or otherwise—that exploit the same resources. Later, ecologists James MacMahon and Charles Hawkins refined the definition of a guild to species that “exploit the same class of environmental resources,” regardless of how they do it.

New ideas

Many new ideas enriched the study of ecology in the late 20th and early 21st centuries. The metapopulation concept was advanced by the Finn Ilkka